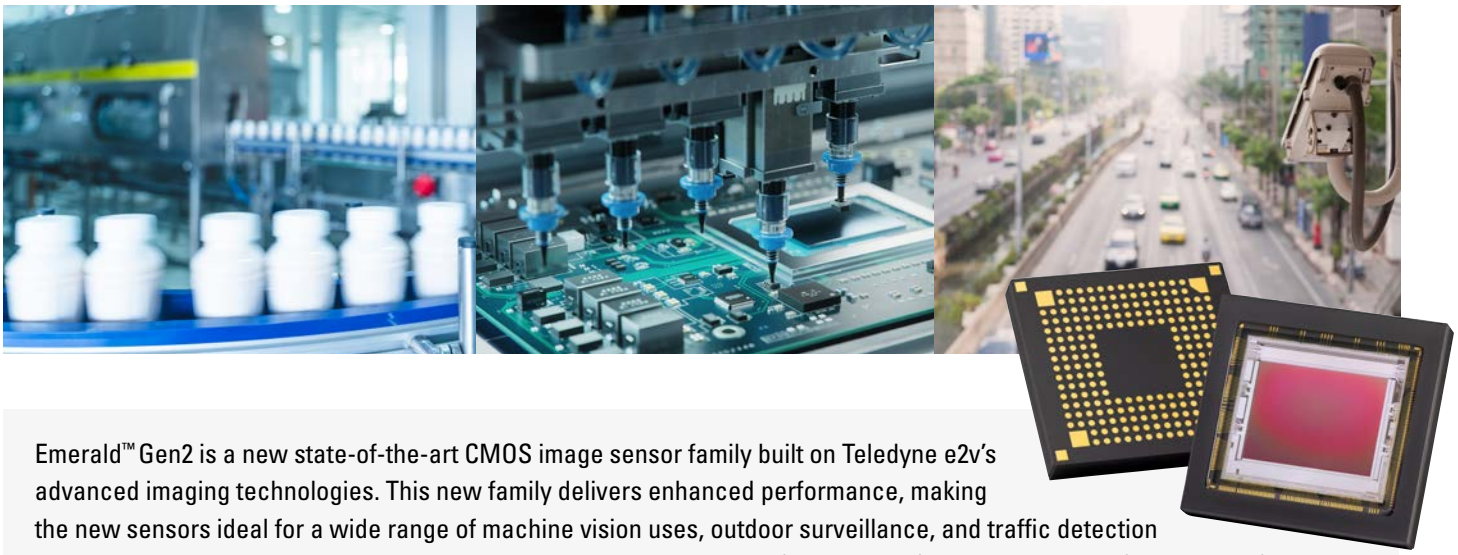
# EMERALD GEN2
# THE NEXT GENERATION OF HIGH-PERFORMANCE GLOBAL SHUTTER CMOS IMAGE SENSORS



Emerald™ Gen2 is a new state-of-the-art CMOS image sensor family built on Teledyne e2v's advanced imaging technologies. This new family delivers enhanced performance, making the new sensors ideal for a wide range of machine vision uses, outdoor surveillance, and traffic detection and monitoring cameras. Emerald Gen2 is available in 8.9-Megapixels (4,096 x 2,160) or 12-Megapixels (4,096 x 3,072), in monochrome or color. There are two-speed grades (standard and high) for demanding applications that require sharp images at very high speeds, such as food sorting, inspection, and Intelligent Traffic Systems.

| SENSOR FEATURES | CUSTOMER BENEFITS |
|---|---|
| High image processing throughput with 100 fps frame rate @ 8.9M resolution | Quick and accurate inspection capabilities |
| Compact 21 x 20 mm² Ceramic Land Grid Array (CLGA) package with centered matrix | Compatible with 1-inch C-mount optics for easy integration and reduced costs |
| A state-of-the-art low noise global shutter | Delivering excellent low-light SNR and the ability to work in high-contrast scenes |
| LVDS and MIPI CSI 2 output | Flexibility in the choice of system architecture |

## Sensor Characteristics

| PARAMETER | EMERALD GEN2-8.9M | EMERALD GEN2-12M |
|---|---|---|
| Resolution – pixels | 4,096 (H) x 2,160 (V) | 4,096 (H) x 3,072 (V) |
| Pixel size – square | 2.8 µm | 2.8 µm |
| Aspect ratio | 17:9 | 4:3 |
| Max frame rate @10bits – FPS (standard speed) | 65[1] | 33[1] |
| Max frame rate @10bits –FPS (high speed) | 100[1] | 70[1] |
| Bit depth | 8/10/12 | 8/10/12 |
| Readout noise – e- | 3[1] | 3[1] |
| Qsat – e- | 7,200[1] | 7,200[1] |
| Dynamic range – dB | 67.5[1] | 67.5[1] |
| SNRmax – dB | 38.6[1] | 38.6[1] |
| Q.E. – %, @550 nm | 65[1] | 65[1] |

[1] Preliminary

### KEY SPECIFICATIONS

- Two resolutions (8.9M & 12M) pin-to-pin compatible
- 2.8 µm CMOS global shutter pixel enabling true CDS
- Up to 65 fps @ full resolution (8.9M) at 10 bits
- Two different speed grades
- Compact ceramic LGA package 20 x 21 mm²
- Two different interfaces (LVDS & MIPI CSI-2)
- 16 LVDS outputs @850 Mbps per lane
- Power consumption: ≤2W @full speed & full resolution

### EMBEDDED FEATURES

- Single Integration Multiple ROI (SIMR): up to 4, non-overlapping and the same integration for each ROI, improving frame rate
- Multi Integration Multiple ROI (MIMR): 2 independent ROI's with possible overlapping and with separated exposure, gain and repetition numbers for each ROI, improving frame rate
- High Dynamic Range mode (dual exposure with MIMR)
- Row sub sampling
- Look-up table
- Improved defective pixel correction
- FPN correction
- Flipping/mirroring
- Image statistics and context output
- Multiple trigger modes

### APPLICATIONS

- Machine vision
- Product inspection and sorting
- Pick and place robot guidance
- Outdoor surveillance
- Intelligent Transportation Systems

### ORDER CODES

| Description | Part Number |
|---|---|
| Emerald Gen2 - 8.9M, Mono, Standard speed | EV2S8M9B-CLM0150-T |
| Emerald Gen2 - 8.9M, Mono, High speed | EV2S8M9B-CLM0350-T |
| Emerald Gen2 - 8.9M, Color, Standard speed | EV2S8M9C-CLM0150-T |
| Emerald Gen2 - 8.9M, Color, High speed | EV2S8M9C-CLM0350-T |
| Emerald Gen2 - 12M, Mono, Standard speed | EV2S12MB-CLM0150-T |
| Emerald Gen2 - 12M, Mono, High speed | EV2S12MB-CLM0350-T |
| Emerald Gen2 - 12M, Color, Standard speed | EV2S12MC-CLM0150-T |
| Emerald Gen2 - 12M, Color, High speed | EV2S12MC-CLM0350-T |

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